

U.S. Inflation, Wage Growth & Purchasing Power Analysis

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Project Type: End-to-End Business Analytics Case Study

Data Source: U.S. Bureau of Labor Statistics (BLS)

Executive Summary

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Project U.S. Inflation, Wage Growth & Purchasing Power Analysis Latest Economic Indicators
----- Latest Year : 2026 Average CPI : 330.08
Inflation Rate : 2.53% Average Hourly Wage : \$37.39 Wage Growth : 2.61% Real Wage Growth : 0.08% Business Insights ----- 1. Latest available year (2026) recorded an inflation rate of 2.53%. 2. Highest inflation occurred in 2022 at 8.00%. 3. Highest wage growth occurred in 2022 at 5.39%. 4. Weakest purchasing power occurred in 2022, when Real Wage Growth fell to -2.42%. Conclusion ----- Overall purchasing power has improved because wages grew faster than inflation.

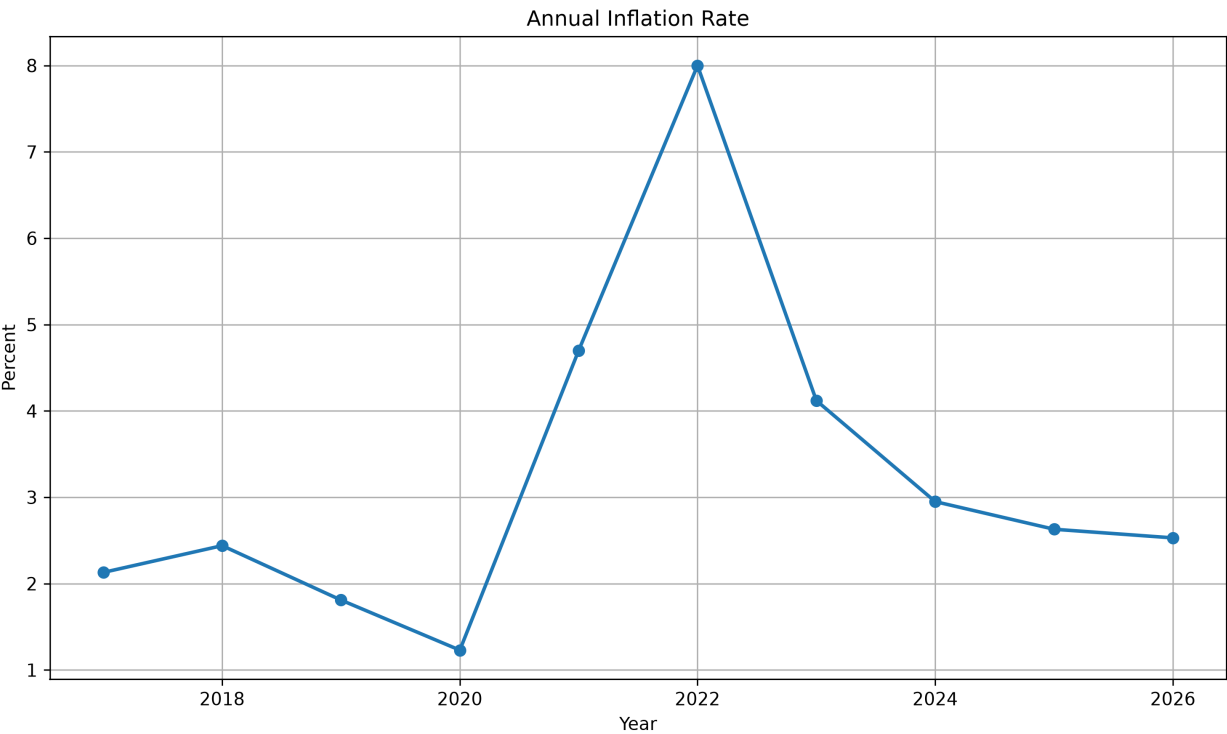
Dashboard KPIs

Metric	Value
Latest Year	2026
Average CPI	330.08
Inflation Rate (%)	2.53%
Average Hourly Wage	\$37.39
Wage Growth (%)	2.61%
Real Wage Growth (%)	0.08%
Employment	158,731
Unemployment Rate (%)	4.30%

Business Insights

{'Latest Economic Conditions': {'Year': 2026, 'Average CPI': np.float64(330.08), 'Inflation Rate': np.float64(2.53), 'Average Hourly Wage': np.float64(37.39), 'Wage Growth': np.float64(2.61), 'Real Wage Growth': np.float64(0.08)}, 'Highest Inflation': {'Year': 2022, 'Inflation Rate': np.float64(8.0)}, 'Highest Wage Growth': {'Year': 2022, 'Wage Growth': np.float64(5.39)}, 'Lowest Real Wage Growth': {'Year': 2022, 'Real Wage Growth': np.float64(-2.42)}, 'Average Inflation': {'Average Inflation': np.float64(3.25)}, 'Average Wage Growth': {'Average Wage Growth': np.float64(3.84)}, 'Purchasing Power Trend': {'Average Real Wage Growth': np.float64(0.6), 'Trend': 'Purchasing power generally improved because wages grew faster than inflation.'}, 'Executive Recommendations': ['Maintain wage growth above inflation to preserve purchasing power.', 'Monitor inflationary pressures closely.', 'Track employment, wages, and inflation together when evaluating the economy.']}

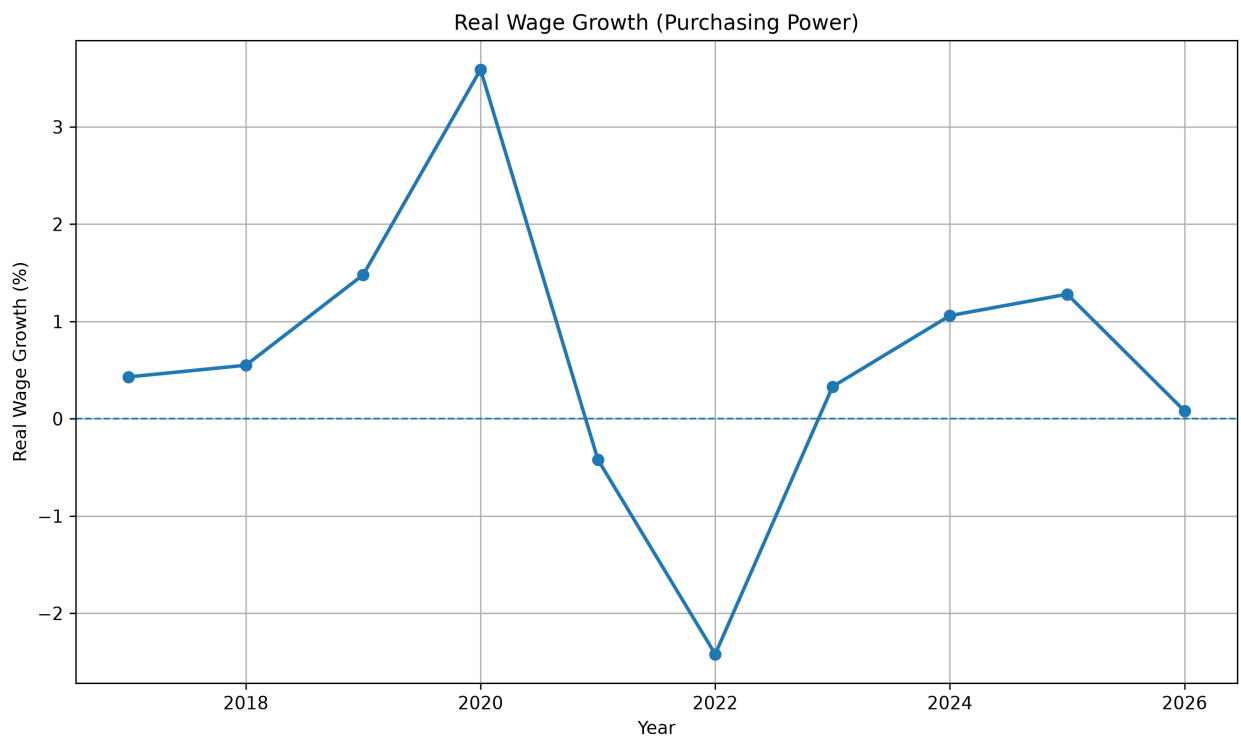
Annual Inflation Trend



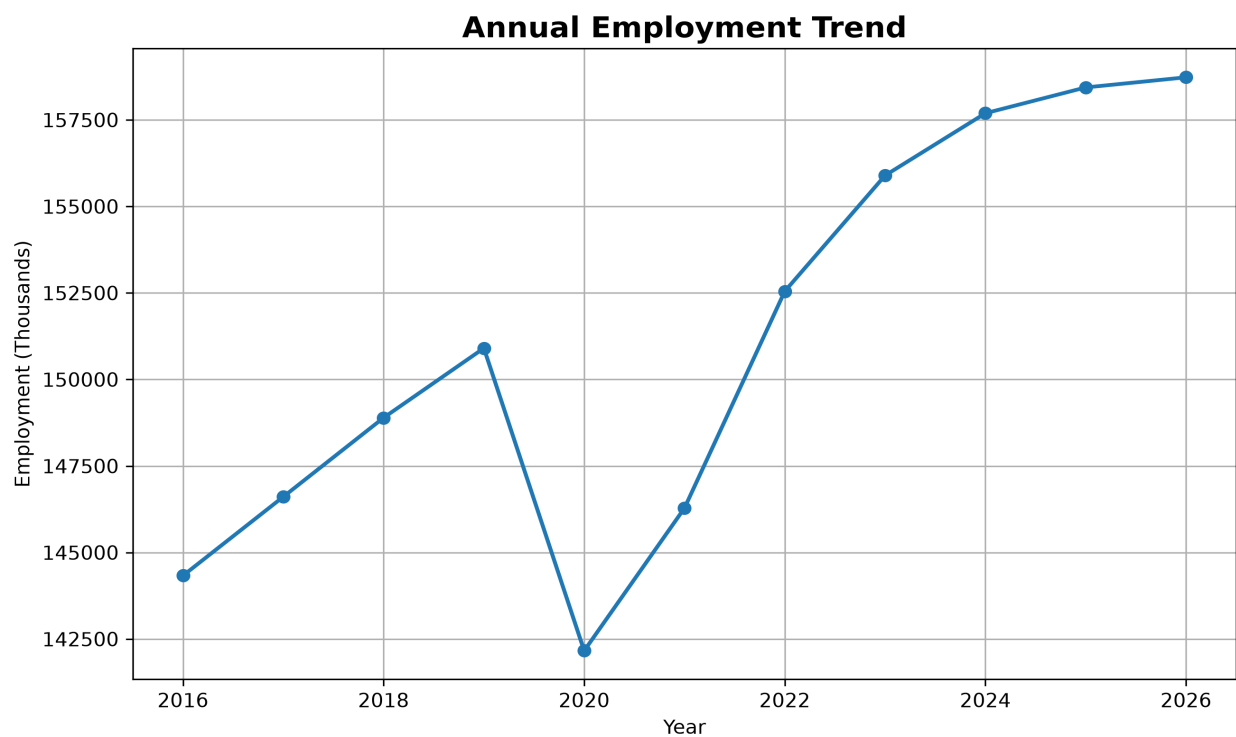
Annual Wage Growth



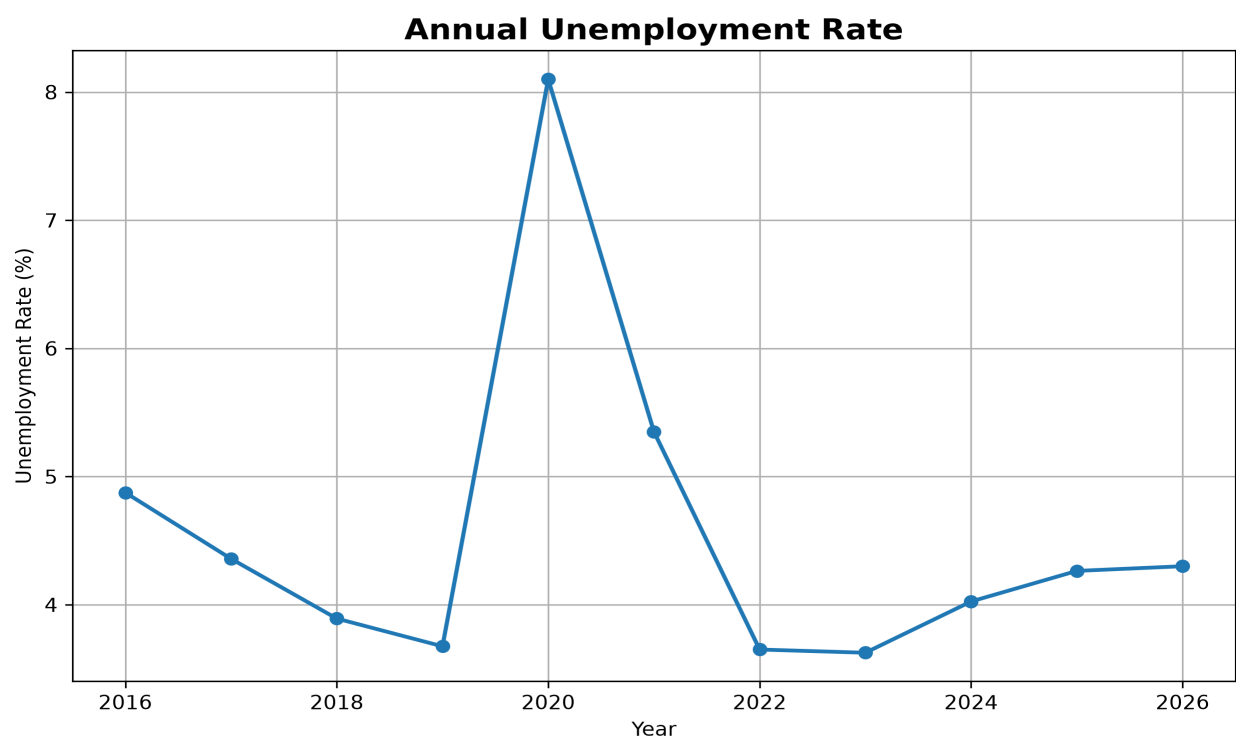
Real Wage Growth



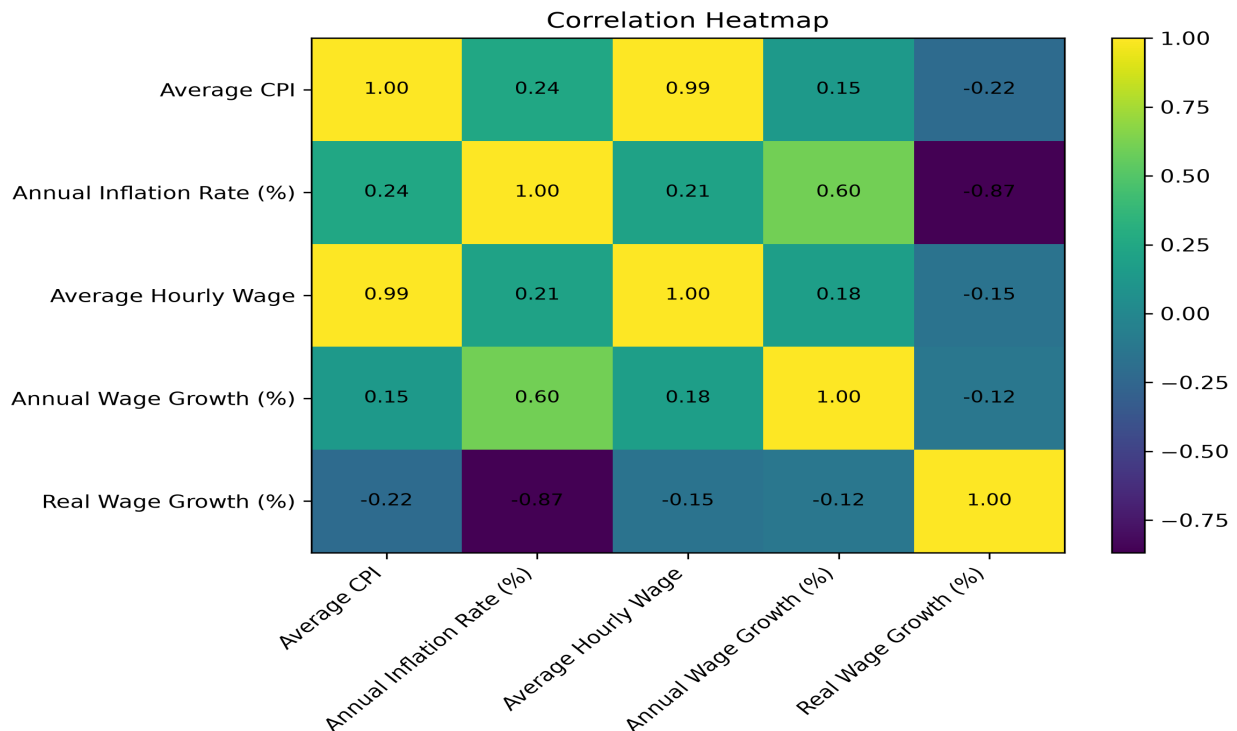
Employment Trend



Unemployment Trend



Correlation Heatmap



Statistical Highlights

- Inflation and Wage Growth were positively correlated during the study period.
- Real Wage Growth became negative during periods of exceptionally high inflation.
- Wage growth generally outpaced inflation over the eleven-year analysis, resulting in gradual improvements in purchasing power.
- Employment continued to trend upward following the COVID-19 recovery while unemployment returned to historically lower levels.

Final Conclusions

This project demonstrates a complete Business Analytics workflow using official U.S. Bureau of Labor Statistics (BLS) data. The analysis integrates ETL, data engineering, feature engineering, statistical analysis, visualization, KPI development and executive reporting into a reproducible analytics pipeline. The results indicate that although inflation surged during 2022, wage growth eventually recovered, allowing purchasing power to improve in subsequent years. Employment continued to strengthen while unemployment stabilized near historically normal levels. This project showcases practical data analytics skills including Python, Pandas, data cleaning, business intelligence reporting, visualization, statistical analysis and executive communication.